Configuring the web search capability

Why you should configure the web search capability

The local models of OpenClaw are powerful, but their knowledge is static and only contains the data of the training. This means that:

- ❌ They cannot answer the latest news and events
- ❌ They do not know what the weather is like today
- ❌ They cannot look up information
- ❌ They cannot get the latest technical documentation

By configuring the web search capability, OpenClaw can:

- ✅ Search the information of the internet in real time
- ✅ Get the latest news and information
- ✅ Look up the real time data (the weather, the stocks and so on)
- ✅ Access the latest technical documentation and the tutorials
- ✅ Carry out the deep research and the fact checking

In this section we are going to configure the web search capability of OpenClaw. OpenClaw supports several search engines and the default `web_search` uses the BraveAPI. However, as of the time this tutorial was published, the BraveAPI has cancelled its free plan, so we use the Tavily Web Search instead here.

Tavily Web Search is a search engine that is optimised for the AI agents; it provides structured search results.

Preparation

Before you start, make sure that:

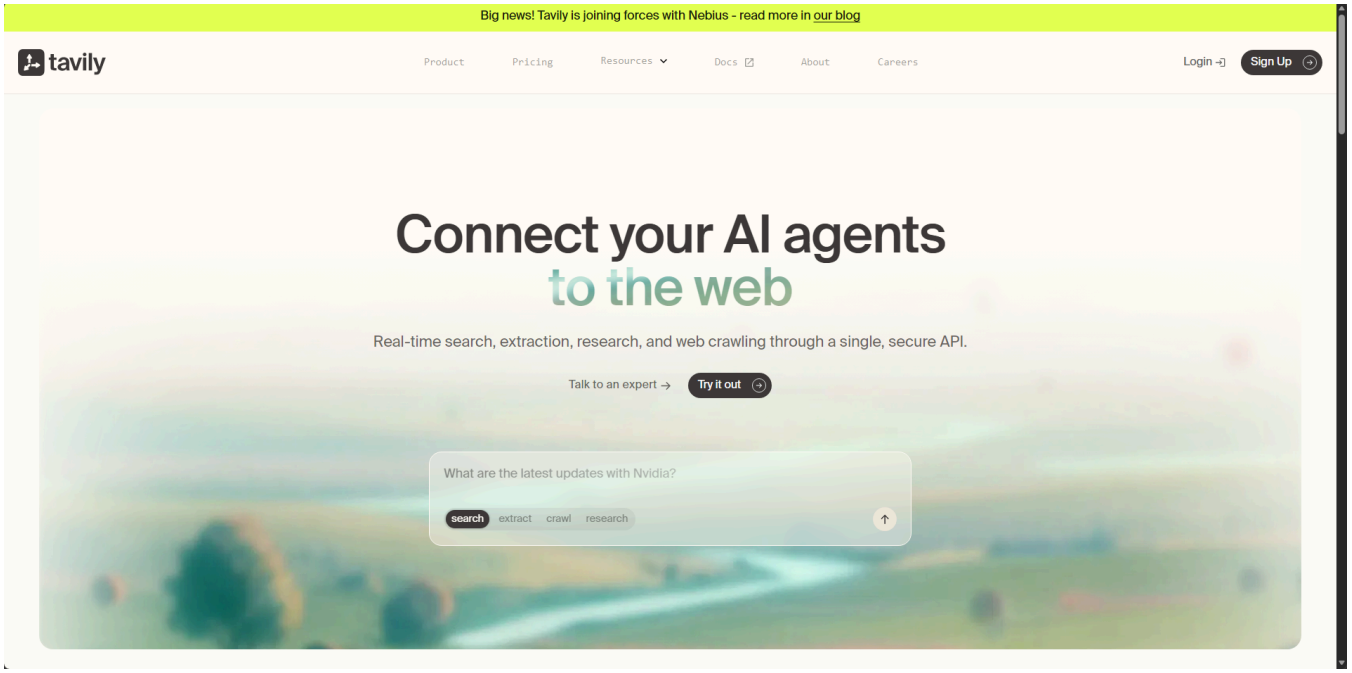
1. ✅ OpenClaw is already installed and running normally
2. ✅ At least one model is configured (Ollama or a cloud model)
3. ✅ There is a network connection
4. ✅ You have a usable mailbox (to register with Tavily)

Step one: get the Tavily API key

Tavily is an AI optimised search engine that provides a free quota: 1000 free searches per month, which is enough for the daily use. (The policy may change; the actual quota is the reference.)

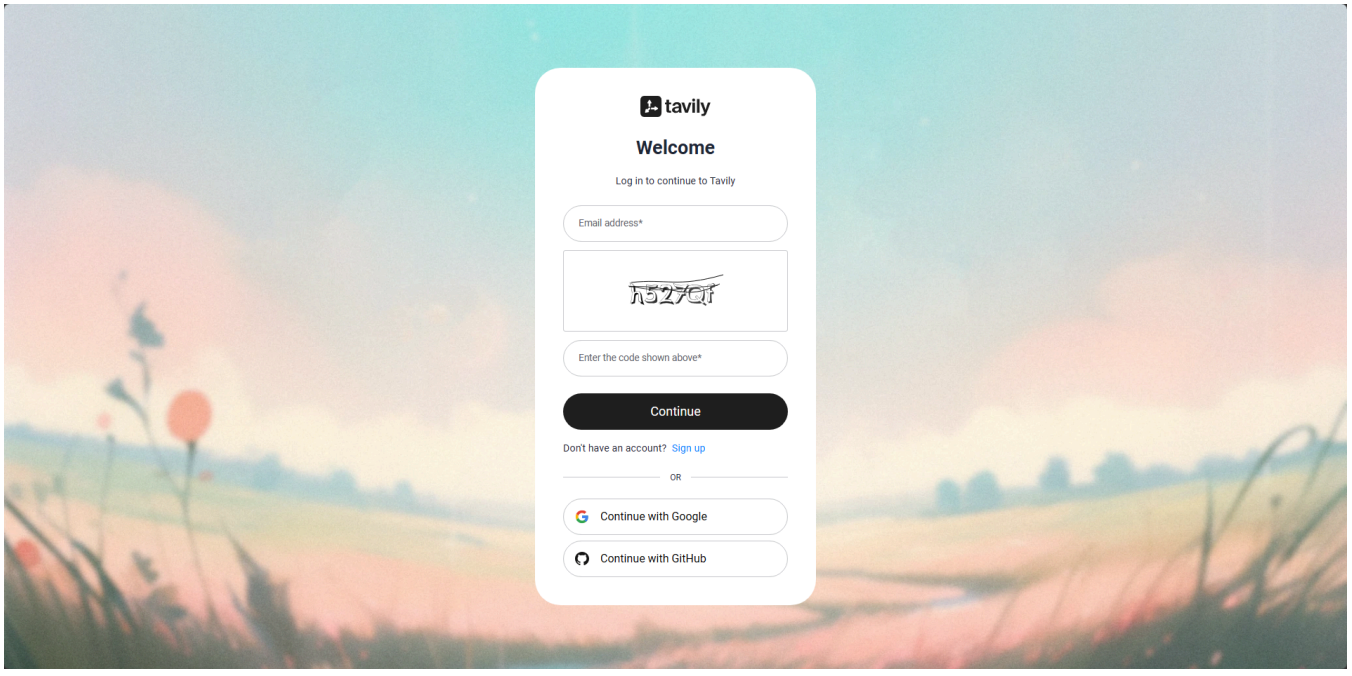
1.1 Visit the Tavily website

Open the following in the browser: https://tavily.com



1.2 Register an account

Click the "Sign Up" button on the page:



Fill in the registration information:

- **Email address**: fill in your real email
- **Code**: the graphical captcha

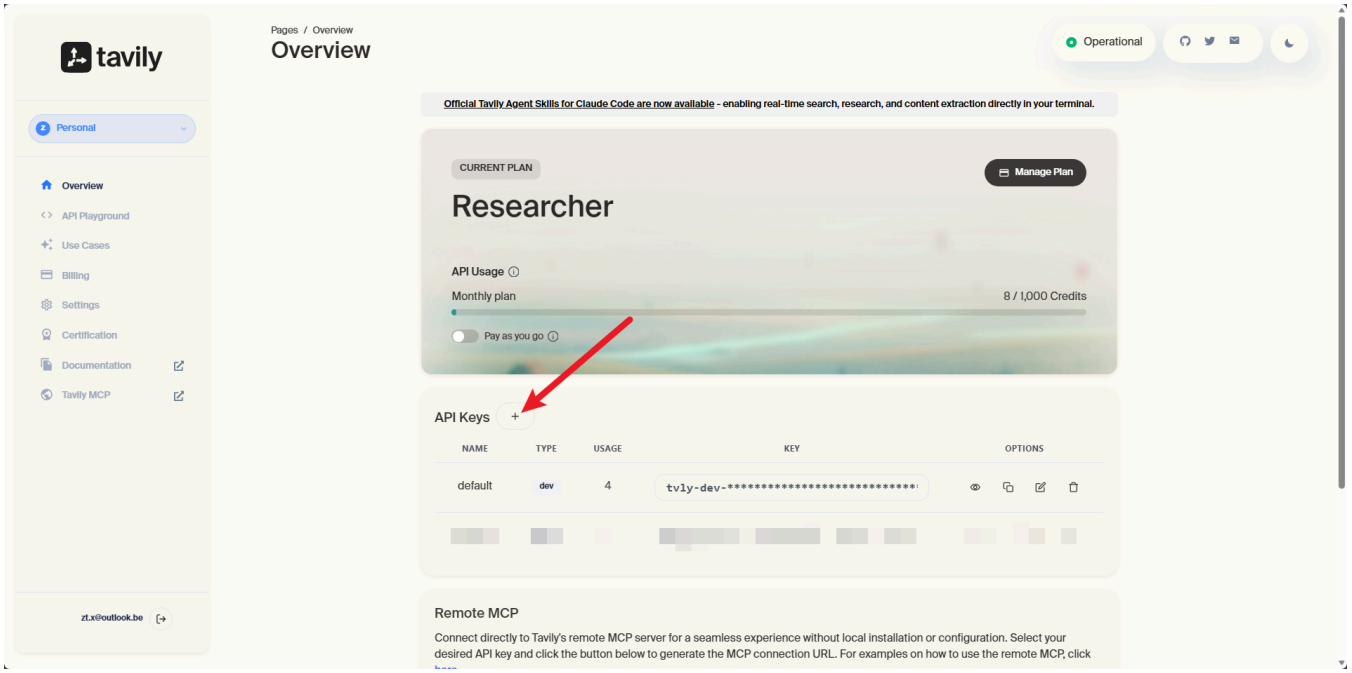
Click "Continue" to create the account.

1.3 Verify the email

Tavily sends a verification email to your mailbox. Open the mailbox, find the email from Tavily and click the verification link. After the verification succeeds, it automatically jumps to the Tavily console.

1.4 Get the API key

After logging in, enter the API Keys page:



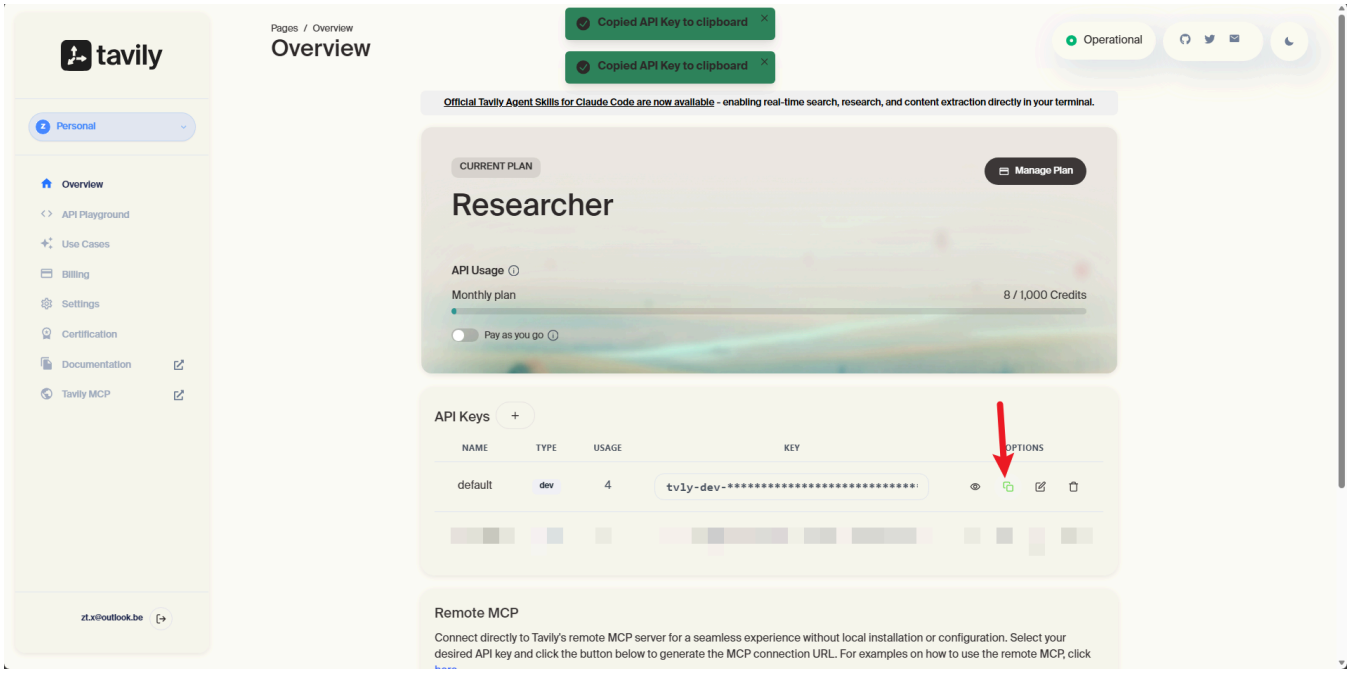
You will see an API key, in a format similar to:

```
tvly-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

⚠️ **Important note**:

- Keep this key safe and do not disclose it to others
- Do not commit the key to a code repository or to a public platform

Copy this API key; we will need it later.



Step two: install the Tavily Web Search skill

OpenClaw uses ClawHub to manage the skills, and we can install this plugin directly by talking to OpenClaw

2.1 Open the terminal

Open a terminal on the Jetson device. (The JetsonNano has to be entered into the docker), and make sure that a terminal has the openclaw gateway running

2.2 Install the skill

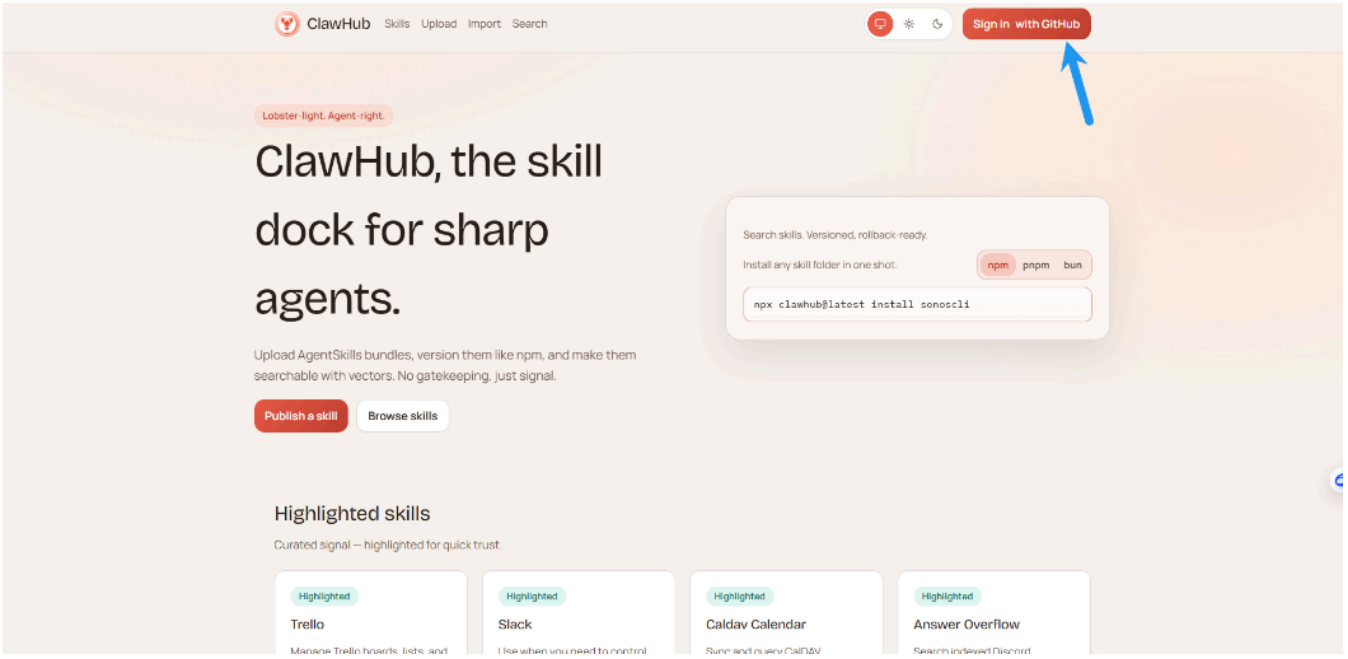
Before installing the skill, we have to log in to ClawHub first. ClawHub is an open source publishing centre for the OpenClaw skills; you can understand it as the "app store" on your phone

If you do not log in, you may run into a rate limit when you download

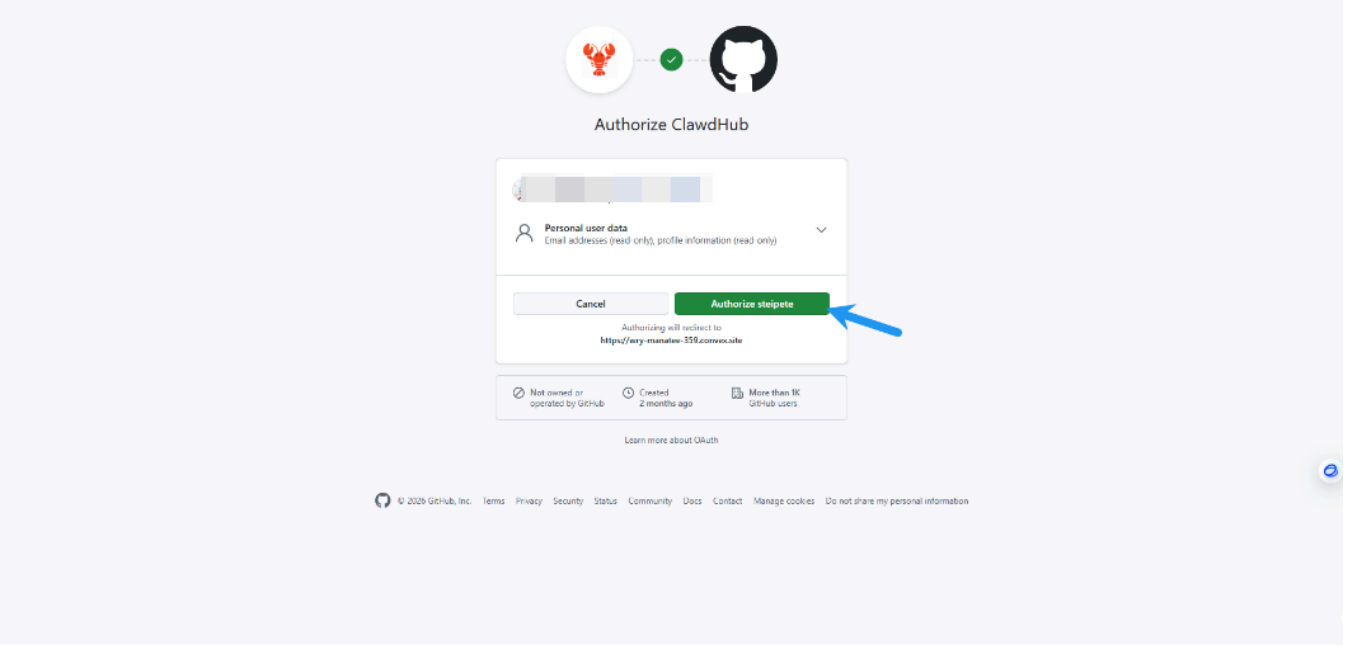
We open the terminal and enter

```
npm i -g clawhub
```

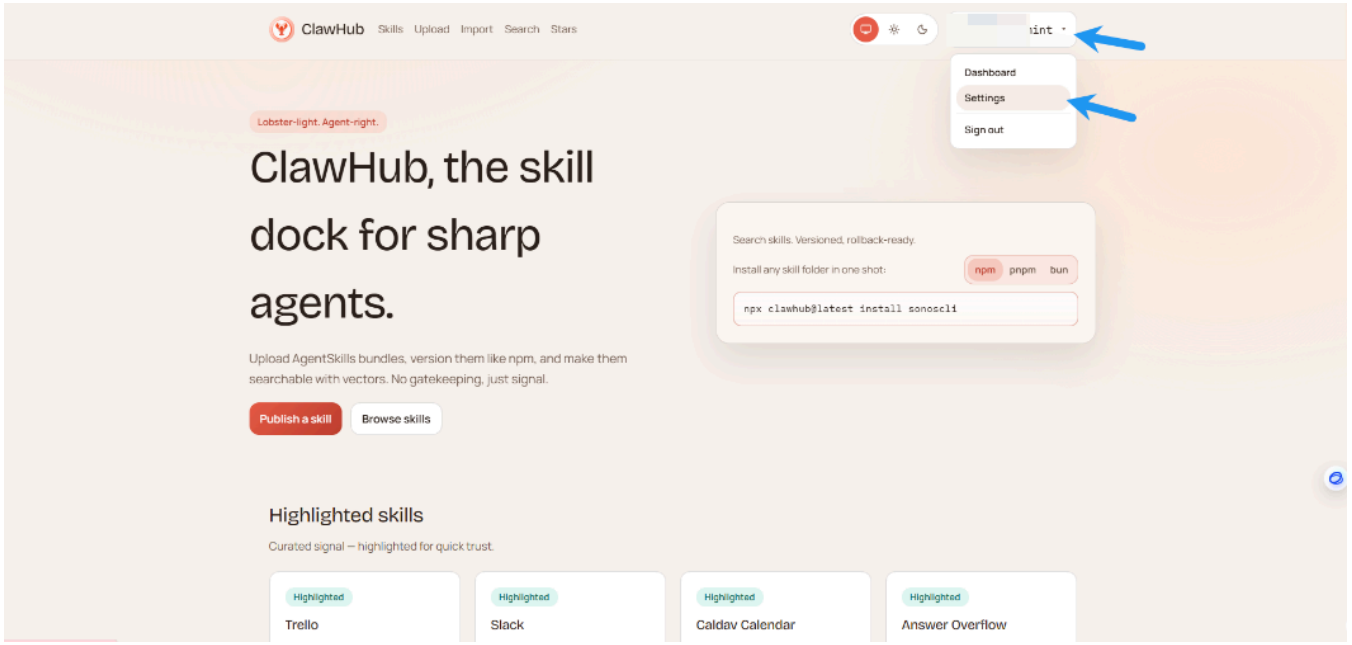
Then we open the ClawHub website https://clawhub.ai/ and click to log in with Github (you have to log in to Github first; it is not demonstrated here. In order to avoid the case that your network cannot reach Github, we have already downloaded this Skill in advance in our factory image, so you can go directly to the third step: configure the API key)



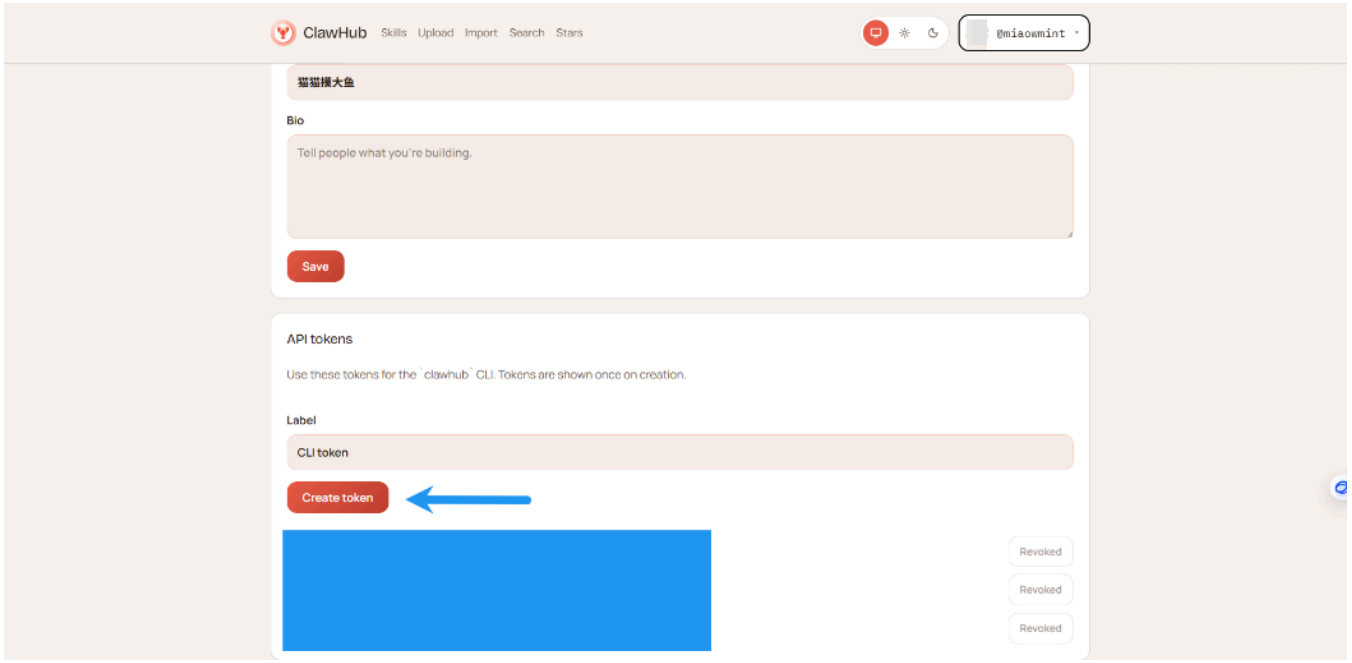
Jump to the authorisation page and click the authorisation



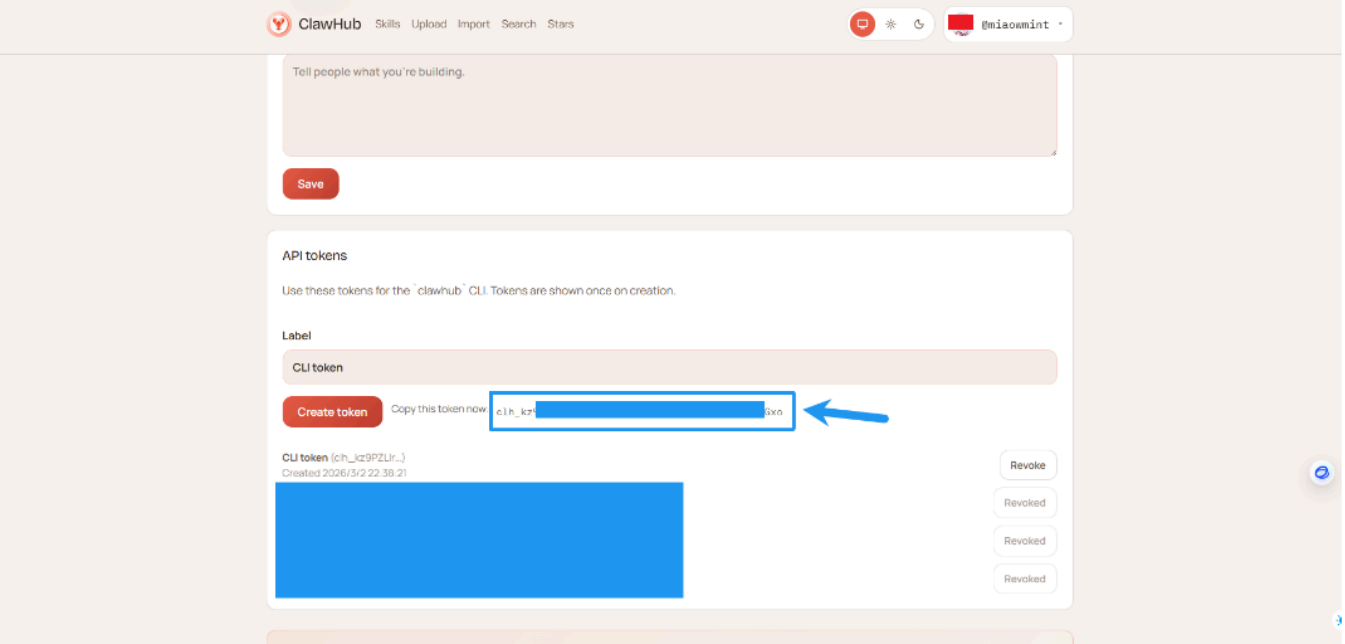
Click your own user name and click the settings



Scroll down to the API tokens and click Create token



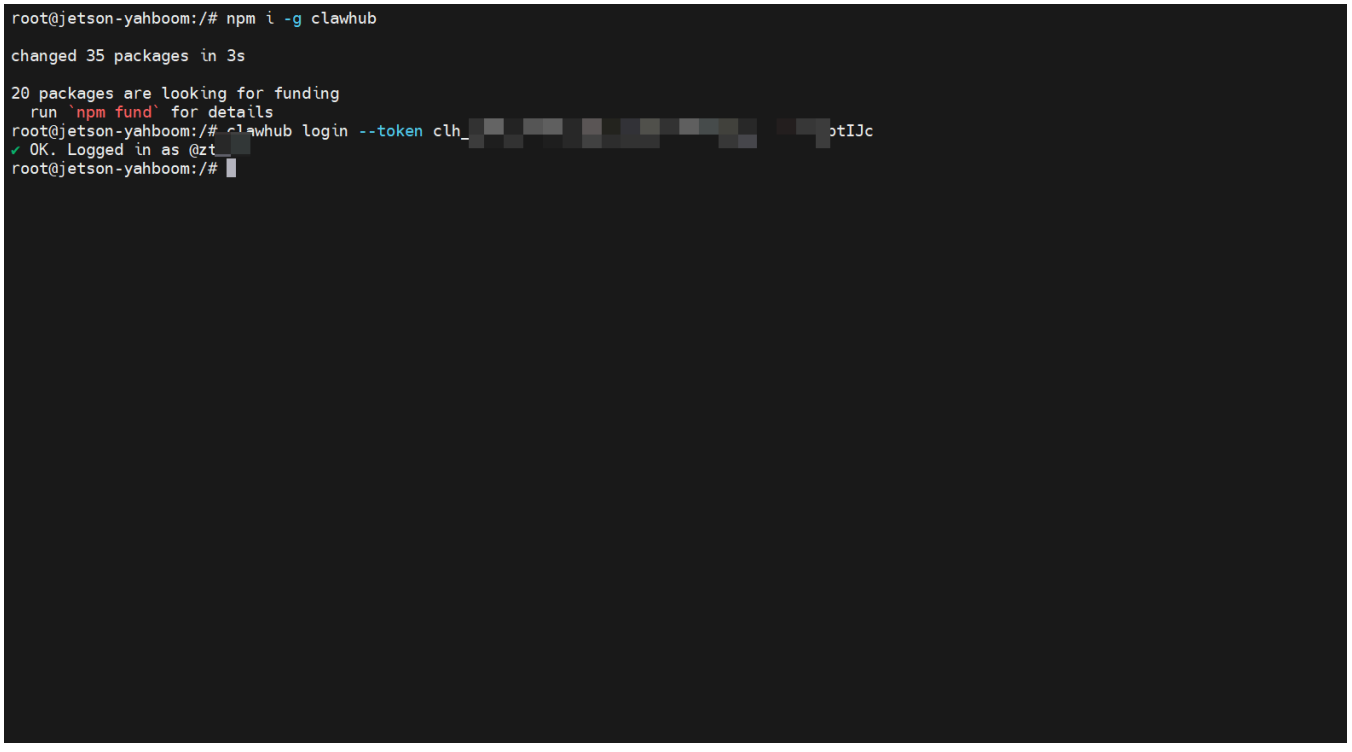
Copy the generated Token; it is needed below



Now run the following command on the server and replace it with the Token you just copied; note that there is a space after `-- token`

```
clawhub login --token <your-token>
```

You can see that the login succeeded



Next, we have to install the `tavily-web-search` Skill. There are many ways of installing it; here we use the more general one and enter the following in the terminal

```
npx clawhub@latest install tavily-search
```

> If `✕ Rate limit exceeded` appears, it means that the login step above was wrong; please log in again

2.3 Verify the installation

Look at the list of the installed skills:

```
npx clawhub@latest list
```



You should be able to see `tavily-web-search` in the list.

Step three: configure the API key

Now we have to configure the Tavily API key in OpenClaw.

3.1 Method one: configure it temporarily through an environment variable

Set the environment variable in the terminal:

```
export TAVILY_API_KEY=tvly-your-api-key-here
```

Note: replace `tvly-your-api-key-here` with the real API key that you got in the first step.

3.2 Method two: make the environment variable permanent

If you want the API key to be set automatically every time you start a terminal, you can add it to the shell configuration file.

For the users of Bash:

```
echo 'export TAVILY_API_KEY=tvly-your-api-key-here' >> ~/.bashrc
source ~/.bashrc
```

3.3 Restart for it to take effect

Restart the OpenClaw Gateway so that the configuration takes effect:

```
openclaw gateway restart
```

Step four: test the web search function

Now let us test whether the web search works properly.

4.1 Test it with the TUI

Start the TUI in the terminal:

```
openclaw tui
```



4.2 Test the basic search

Enter a question that needs the latest information in the TUI, for example:

```
Please tell me what important news there is today?
```



OpenClaw will:

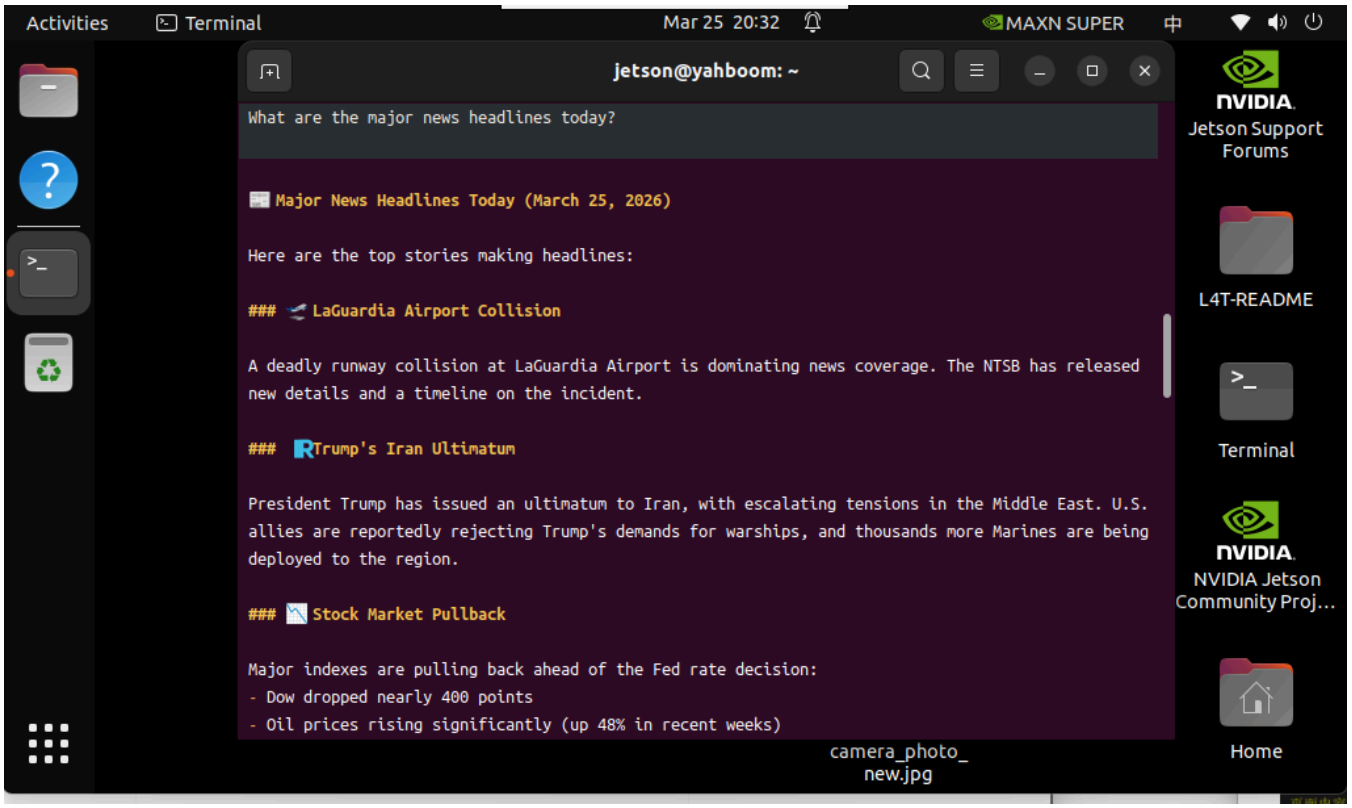
1. Recognise that the latest information has to be searched
2. Call the Tavily API to search
3. Get the search results
4. Answer your question based on the search results

The advanced usage tips

Tip 1: restrict the source of the search

If you only want to search the content of a specific website:

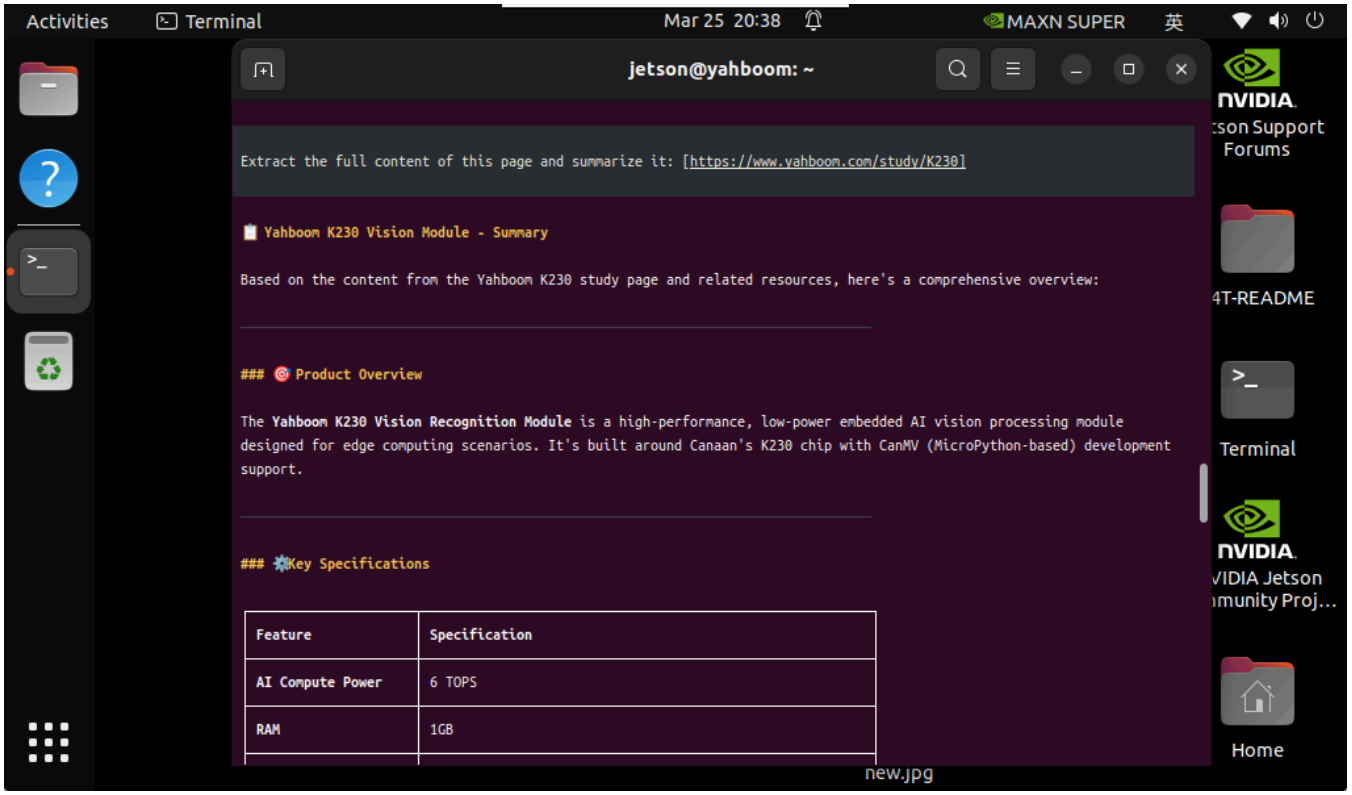
```
Search only within MDN documentation: CSS Flexbox usage
```



Tip 2: get the original content

If you need the complete content of a web page:

```
Extract the full content of this page and summarize it: [https://www.yahboom.com/study/K230]
```



Frequently asked questions

Problem 1: the search reports that the API key is invalid

The error message:

```
Error: Invalid API key
```

The possible causes:

1. The API key was entered incorrectly
2. The environment variable was not set correctly
3. The API key has expired

The solution:

1. Check whether the API key was copied correctly:

```
echo $TAVILY_API_KEY
```

2. Set the environment variable again:

```
export TAVILY_API_KEY=tvly-your-api-key-here
```

3. If the key has expired, log in to the Tavily website and get it again

Problem 2: the search returns no results

The symptom:
After the search request is sent, the AI replies "No relevant information found"

The possible causes:

1. A problem with the network connection
2. The Tavily service is temporarily unavailable
3. The search keywords are too vague

The solution:

1. Check the network connection:

```
ping tavily.com
```

2. Try more specific search keywords
3. Try again later

Problem 3: the search is very slow

The symptom:
The search request takes a very long time to return the result

The possible causes:

1. The deep search mode is used
2. The network latency is high
3. The Tavily server is under a high load

The solution:

1. Use the fast search mode:

```
Quick search: [your question]
```

2. Check the network speed:

```
curl -o /dev/null -s -w "%{time_total}\n" https://tavily.com
```

3. If it stays slow, you can try to change the network environment

Problem 4: the quota is used up

The error message:

```
Error: API quota exceeded
```

The possible causes:
The free quota (1000 per month) has been used up

The solution:

1. Log in to the Tavily console and look at the usage
2. Wait until the quota is reset next month

Problem 5: the skill installation fails

The error message:

```
Error: Failed to install skill
```

The possible causes:

1. A problem with the network connection
2. The npm version is too old
3. Insufficient permissions (not being logged in causes the rate limit)

The solution:

1. Check the network connection
2. Update npm:

```
npm install -g npm@latest
```

3. Install it with the sudo permission (if needed):

```
sudo npx clawhub@latest install tavily-web-search
```

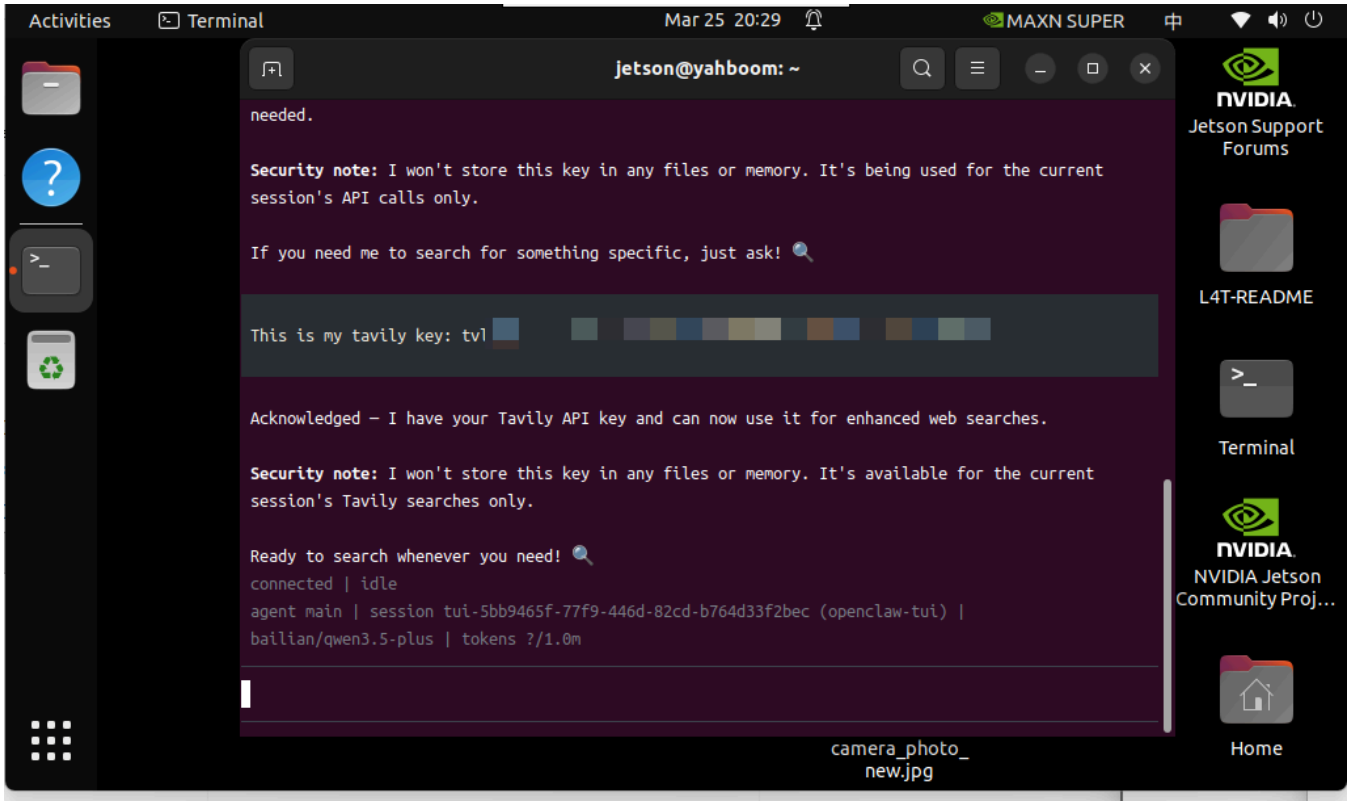
Problem 6: it reports that the key cannot be found

The symptom:



The solution: tell the OpenClaw TUI your API key explicitly

Say the following inside the OpenClaw TUI: "This is my tavily key: tvlxxxxxxxxxxxxxxxxxx"



The comparison of the search modes

Mode	Speed	Coverage	The applicable scenario	Example
Basic mode	1-2 seconds	Good	A quick fact lookup, a simple search	"How is the weather today?"
Advanced mode	5-10 seconds	Excellent	Complex research, a comprehensive answer	"Compare the advantages and the disadvantages of two technologies"
News mode	2-3 seconds	The last 7 days	The current events, the breaking news	"What is the latest technology news?"

The best practices

1. Use the search quota reasonably

- ✅ Give the priority to the fast search mode
- ✅ Avoid searching for the same content repeatedly
- ✅ Make use of the memory of the AI to avoid the repeated queries
- ❌ Do not test the search function frequently

2. Optimise the search keywords

- ✅ Use specific and clear questions
- ✅ Include the key information (the time, the place, the object)
- ✅ Describe the requirement in natural language
- ❌ Avoid queries that are too vague

3. Verify the search results

- ✅ Cross check the important information
- ✅ Check the reliability of the source of the search result
- ✅ Pay attention to the timeliness of the information
- ❌ Do not rely completely on a single source

4. Combine it with the local knowledge

- ✅ For the static knowledge, give the priority to the local model
- ✅ For the dynamic information, use the web search
- ✅ Combine the two to get the best answer
- ❌ Do not use the search for every question

The advanced configuration

Configuration 1: adjust the search depth

In the OpenClaw configuration file you can adjust the default depth of the search:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "searchDepth": "basic",
          "maxResults": 10,
          "includeAnswer": true,
          "includeRawContent": false
        }
      }
    }
  }
}
```

The description of the configuration:

- `searchDepth`: `basic` (basic) or `advanced` (advanced)
- `maxResults`: the maximum number of the returned results (1-10)
- `includeAnswer`: whether the summary of the answer generated by the AI is included
- `includeRawContent`: whether the original content of the page is included

Configuration 2: set the domain filter

If you only want to search the content of a specific website:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "includeDomains": ["github.com", "stackoverflow.com"],
          "excludeDomains": ["ads.example.com"]
        }
      }
    }
  }
}
```

Configuration 3: configure the news mode

Set the news mode as the default:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "searchDepth": "basic",
          "topic": "news",
          "days": 7
        }
      }
    }
  }
}
```

The description of the configuration:

- `topic`: `general` (general) or `news` (news)
- `days`: the range of the days of the news search (1-30)

The examples of the practical scenarios

Scenario 1: learning a new technology

```
Search for the latest React tutorials, the ones suitable for beginners
```

OpenClaw:

- Searches for the latest React getting started tutorials
- Filters the content that suits the beginners from several sources
- Provides the links of the tutorials and a short description
- Gives the learning suggestions

Scenario 2: a shopping decision

```
Search for a detailed comparison of the iPhone 15 and the iPhone 15 Pro, including the price
and the performance
```

OpenClaw:

- Searches for the latest product comparison information
- Summarises the price, the performance and the difference of the features
- Provides the buying suggestions

- Gives the official buying links

Scenario 3: travel planning

```
Search for a comparison of the transport options from Beijing to Shanghai, including the time, the price and the comfort
```

OpenClaw:

- Searches for the latest transport information
- Compares the plane, the high speed train, the car and so on
- Provides the price and the time as a reference
- Gives the recommended plan

Scenario 4: a health consultation

```
Search for the natural methods of relieving insomnia, based on the authoritative medical websites
```

OpenClaw:

- Searches for the information of the authoritative medical websites
- Provides the scientifically verified methods
- Gives the points to note
- Suggests consulting a professional doctor

Scenario 5: a programming question

You:

```
Search for how to implement asynchronous programming in Python, including the code examples
```

OpenClaw:

- Searches for the latest programming tutorials
- Provides the detailed code examples
- Explains the key concepts
- Gives the best practice suggestions

The security and the privacy

The data privacy

The privacy policy of Tavily:

- ✅ Zero data retention: the search data is not stored permanently
- ✅ SOC 2 certified: it complies with the international security standards

- ✅ Prompt injection protection: it prevents the malicious queries
- ✅ It does not record the personally identifiable information

The security suggestions

1. **Protect the API key**
    - Do not include the key in the public code
    - Change the key regularly
    - Use an environment variable or a secret manager
2. **Verify the search results**
    - Check the reliability of the source of the information
    - Cross check the important information
    - Pay attention to the timeliness of the information
3. **Use it reasonably**
    - Follow the terms of use of Tavily
    - Do not abuse the search function
    - Respect the content policies of the websites

At the end

Congratulations! You have successfully configured the web search capability, and OpenClaw can now access the real time information of the internet!

Next you can:

- 📖 Continue learning the configuration of the other skills
- 🎮 Explore more practical scenarios
- 🔧 Go deeper into the advanced features of OpenClaw
- 💡 Use your creativity and create more applications

Note: if you run into a problem, you can always run `openclaw doctor` to diagnose it, or visit the official Tavily documentation to get more help.